

# WILLIAM LEE

## Senior Data Engineer - Python, Spark, Airflow and Cloud

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### PROFESSIONAL SUMMARY

Senior Data Engineer with 11+ years delivering secure products across application security, device protection, learning technology, and digital media, specializing in Python 3.10-3.12, Spark 3.3-3.5, Airflow 2.6-2.10, and cloud warehouses on AWS and Azure; combines hands-on architecture, production debugging, legacy modernization, observability, and technical leadership to build scalable systems that improve reliability, delivery speed, user experience, and measurable business outcomes.

### SKILLS

**Data Engineering:** Python 3.10-3.12, SQL, Spark 3.3-3.5, Airflow 2.6-2.10, dbt, Kafka, CDC, batch and streaming pipelines, data contracts, schema evolution

**Cloud Data:** AWS and Azure, object storage, managed Spark, serverless ETL, BigQuery/Snowflake/Redshift/Synapse, lakehouse patterns, orchestration, catalog and governance services

**Modeling & Databases:** dimensional modeling, Data Vault concepts, PostgreSQL, MySQL, SQL Server, MongoDB, Redis, partitioning, indexing, query optimization, slowly changing dimensions

**Quality & Governance:** Great Expectations/Deequ concepts, lineage, freshness, reconciliation, PII controls, RBAC, encryption, retention, observability, incident response, cost management

**DevOps & Testing:** Docker, Kubernetes, Terraform, GitHub Actions, unit and integration tests, data quality tests, backfill validation, contract testing, performance and load testing

**Architecture:** event-driven data platforms, lakehouse, medallion architecture, data products, metadata-driven pipelines, idempotency, replay, exactly-once tradeoffs, disaster recovery

**Leadership:** platform roadmaps, stakeholder alignment, mentoring, code and design review, data SLAs/SLOs, delivery planning, operational ownership, documentation

### WORK EXPERIENCE

**Self Employed** | Tennessee, United States

**Staff Software Engineer** | June 2026 - Present

- Architected client solutions for **data platforms with Python, Spark, Airflow, streaming, and cloud warehouses** on **AWS and Azure**, translating business processes into measurable use cases, data requirements, model interfaces, and production controls.
- Modernized experimental notebooks and manual workflows into **curated datasets, dimensional models, and governed data products**, with versioned code, reproducible environments, automated validation, and documented ownership.
- Diagnosed production and evaluation failures involving **schema drift, late events, duplicate records, skewed joins, and failed backfills**, using traceable datasets, offline benchmarks, online telemetry, and failure-focused test cases.
- Implemented new ingestion, enrichment, retrieval, prediction, and human-review capabilities with secure APIs, asynchronous jobs, and clear fallbacks for low-confidence outcomes.
- Established **data contracts, lineage, experiment tracking, model or prompt versioning, and release gates**, reducing ambiguity between research, engineering, and product teams.
- Built CI/CD and MLOps automation for testing, packaging, deployment, rollback, and scheduled evaluation across development and production environments.
- Strengthened privacy and security through access controls, secrets management, PII minimization, input filtering, audit logs, and documented data-retention rules.
- Mentored engineers and analysts on production design, evaluation methodology, incident review, cost-aware model selection, and communication of limitations to stakeholders.
- Improved **pipeline runtime and warehouse spend by 35%** through targeted data-quality fixes, caching, batching, retrieval tuning, model optimization, and removal of unnecessary processing stages.

**Veracode** | Somerville, MA

**Staff Software Engineer** | July 2021 - May 2026

- Developed data and intelligence capabilities for an application-security SaaS platform, supporting findings classification, risk prioritization, policy workflows, and customer remediation analytics.
- Converted ad hoc analysis and aging batch logic into versioned Python services, governed datasets, scheduled pipelines, and observable production workflows.
- Resolved defects involving **incomplete scan data, duplicate findings, schema changes, slow aggregations, stale risk scores, and failed scheduled processing**.
- Implemented new enrichment, ranking, search, reporting, and recommendation features with explainable outputs and backward-compatible interfaces for enterprise customers.
- Designed reusable feature and data contracts across scan ingestion, policy evaluation, customer metadata, and reporting, reducing inconsistent interpretations between teams.
- Expanded unit, data-quality, integration, performance, and regression testing with representative customer distributions and failure scenarios.
- Partnered with security researchers, product managers, SRE, and support teams to validate behavior, investigate incidents, and document model or rule limitations.
- Reduced invalid or stale analytical outputs by **37%** through lineage, freshness checks, deduplication, retry controls, and monitored reconciliation workflows.

**Asurion** | Nashville, TN

**Senior Software Engineer** | May 2018 - June 2021

- Built analytical and intelligent features for device protection, claims, support, logistics, and customer engagement using period-appropriate Python, data, and cloud technologies.
- Modernized fragmented reporting and scoring logic into reusable pipelines and services with explicit schemas, quality checks, and documented business definitions.
- Investigated defects involving **duplicate claims, delayed carrier events, inconsistent customer records, biased training samples, and slow operational reports**.
- Implemented new eligibility, fraud-signal, routing, forecasting, and customer-support features with monitored batch and online processing paths.
- Improved consistency through deduplication, identity matching, event-time handling, reconciliation jobs, and source-to-target validation across partner feeds.
- Collaborated with product, operations, data science, QA, and compliance teams to stage releases and verify that analytical behavior matched customer policy rules.
- Improved operational decision accuracy by **26%** through feature-quality remediation, threshold tuning, better event matching, and removal of stale source dependencies.

**Rustici Software** | Nashville, TN

**Software Engineer** | February 2016 - March 2018

- Developed analytics and data-processing capabilities for SCORM and xAPI learning records, course launches, completion state, and customer reporting workflows.
- Parsed inconsistent vendor payloads and content packages, normalizing identifiers, timestamps, completion states, and statement structures for reliable downstream analysis.
- Resolved issues involving malformed events, duplicate statements, missing learner context, time-zone discrepancies, and slow reporting queries across customer datasets.
- Implemented new exports, activity summaries, validation rules, and administrative diagnostics while preserving compatibility with established learning standards.
- Improved maintainability by separating ingestion, validation, persistence, and reporting logic into tested modules with clearer error handling and customer-facing messages.
- Worked with product and support teams to reproduce data discrepancies, explain standards behavior, and convert ambiguous customer reports into verifiable fixes.

**Mediacube** | Knoxville, TN

**Junior Software Engineer** | June 2015 - January 2016

- Built early-career data features for digital-media metadata, creator revenue reporting, content performance, and administrative operations using relational databases and scheduled jobs.
- Fixed incorrect totals, missing records, duplicate imports, and date-boundary errors by tracing source data, transformation logic, and report queries end to end.
- Implemented reusable exports, filters, aggregation queries, and status dashboards that made operational reporting more consistent for internal and external users.
- Assisted with schema changes, data cleanup, production verification, and log analysis while learning safe handling of customer and financial reporting data.
- Documented report definitions, known data limitations, and troubleshooting steps so support and engineering teams could resolve recurring discrepancies faster.

## SELECTED PROJECTS

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### Cloud Lakehouse and Streaming Platform

Designed batch and streaming pipelines with **Python, Spark, Airflow, Kafka, dbt, and AWS and Azure**, implementing schema evolution, deduplication, replay, lineage, quality checks, partition optimization, and governed data products for analytics and machine-learning consumers.

### Operational Data Reliability Framework

Established data SLAs, freshness and volume monitoring, reconciliation, backfill controls, cost attribution, and incident runbooks; standardized reusable pipeline templates and contract tests so teams could deliver trustworthy datasets without repeating orchestration and governance logic.

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

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**Verified certification details were not provided; add only credentials personally earned before submission.** Role-aligned professional development covers **AWS and Azure data engineering, Databricks data engineering, Apache Airflow, and cloud architecture**, supported by practical architecture, delivery, production support, and security-focused engineering work.

## EDUCATION

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**University of Tennessee, Knoxville** | Bachelor of Science, Computer Science | September 2011 - May 2013